

Particle Physics — Decay Rates

IMAPP lecture notes

Spring 2026

1 Exponential decay

If $N(t)$ particles are present, a constant decay rate gives

$$dN = -\Gamma N dt, \quad N(t) = N(0)e^{-\Gamma t}, \quad \tau = \Gamma^{-1}.$$

The width Γ is the natural width associated with the finite lifetime.

2 Decay width

For a decay into a set of final states, the differential width is written schematically as

$$d\Gamma = \frac{1}{2E_A} |\mathcal{M}|^2 d\Phi_n,$$

where $\mathcal{M}$ is the Lorentz-invariant matrix element and $d\Phi_n$ is the Lorentz-invariant n -body phase-space measure. For two final particles,

$$d\Phi_2 = (2\pi)^4 \delta^{(4)}\left(p_A - \sum_i p_i\right) \prod_{i=1}^2 \frac{d^3\mathbf{p}_i}{(2\pi)^3 2E_i}.$$

The total width is obtained by integrating over all allowed final-state momenta.

3 Cross-sections

The interaction rate is the incident flux multiplied by the number of target particles and the cross-section:

$$\text{rate} = \text{flux} \times N_{\text{target}} \times \sigma.$$

The cross-section has dimensions of area; the barn is a standard unit, $1 \text{ b} = 10^{-28} \text{ m}^2$.

For two incoming particles, the Lorentz-invariant flux can be expressed in terms of their four-momenta. The invariant matrix element and phase-space factors together ensure that the calculated rate is frame independent.

Summary

Decay widths describe the probability per unit time for an unstable state to decay. Cross-sections describe interaction probabilities per incident flux. Both are built from invariant amplitudes and Lorentz-invariant phase space.